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## Enhancing growth, yield, and nutritional value of *Capsicum Annuum*: Evaluating micronutrient efficiency and varietal performance

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### ABSTRACT

This study investigates the effects of chili variety, micronutrient application efficiency, and their interaction on the growth, yield, and quality attributes of chili (*Capsicum* spp.) over two consecutive years. The experiment was conducted in a split-plot design with three replications. Three chili varieties (V1=Sikarpuri, V2=Sakata-653, V3=Picnic) and three micronutrient application methods (M1: Broadcasting, M2: Spraying, M3: 50 % Broadcasting + 50 % Spraying) and their interactions were examined. For this, the variety was assigned in the main plot and micronutrients in the sub-plot. The results revealed that the interaction of variety and micronutrient application efficiency in V3M3 treatment demonstrated significant differences for growth characteristics, including increased plant height, number of leaves, earlier flowering, and higher fruit count. Yield parameters such as fruit length, diameter, fresh weight, dry weight, and seed weight were significantly enhanced by the interaction of variety and micronutrient application efficiency. The highest fruit length (13.25 cm) and seed weight (0.39 g) were observed in V3M3 treatments. Fresh fruit weight per plant peaked at 193.37 g in V3M3, while the highest fresh fruit yield per hectare (9.72 t/ha) was obtained from the same treatment. V3M3 produced the highest vitamin C (96.101 mg/100 g) and sugar content (1.624 %). The interaction of V3M3 yielded the highest protein content (1.085 %) and ash percentage (1.58 %). Conversely, the lowest quality values were observed in the V2M1 and V1M1 combinations. These results imply that the right mix of micronutrient treatments and variety increased growth, yield, and quality characteristics, offering stakeholders and policymakers important information for better chili farming practices.

### 1. Introduction

Chili (*Capsicum* spp.) is one of the most significant spice crops, which is grown in temperate, tropical, and subtropical climates worldwide (Zhou et al., 2024). Chilies are native to South America and are members of the Solanaceae family of nightshades [1]. The demand for pepper fruits is increasing day by day due to their pungency, delicious flavour, and medicinal value (Zongo et al., 2025). Vitamin C and pro-vitamin A are abundant in chilies, especially red ones [2]. Nutritional content can

vary based on production conditions, genotype, and fruit ripeness. 100 g of fresh hot peppers include around 88 g of water, 40 kcal of calories, and other vitamins and minerals. On the other hand, dried peppers have distinct nutritional profiles, with 13.4 % moisture, 11.9 % fat, 12.8 % protein, 56.2 % carbs, and 22.5 % fiber content [3]. Additionally, chili has antioxidants, antimutagenic, and hypocholesterolemia effects [4]. Globally, chili serves as a source of cash for farmers, reducing poverty and improving women's social status [5]. In Bangladesh, although chili is one of the most important crops; insect-pest infestation, inadequate nutrient management, lack of high-yielding variety, and unfriendly

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| Abbreviation: |                          |
|---------------|--------------------------|
| DAT           | days after transplanting |

weather are the major barriers to getting optimum production ([6]; Shabuj, 2023). The global chili (dry and green) yield in 2023 (FAO STAT,2023) was 21778 kg/ha, whereas Bangladesh experienced a relatively low amount of yield (6810 kg/ha) (BBS,2023).

Micronutrient deficiencies in chili production often limit growth, reduce yield, and compromise fruit quality. ([7,8], Magalhães et al., 2023). One promising approach involves the strategic application of micronutrients, which play a crucial role in physiological and metabolic processes essential for plant growth and development [9–11]. Proper application methods can enhance the efficiency of micronutrients (Batool, 2023, [12]), the effectiveness depends on the type of micronutrient and the chili variety. Despite their importance, systematic research on how different micronutrient application methods influence chili growth and productivity remains limited.

Understanding the interaction between application strategies and varietal responses is vital for developing efficient production protocols. In recent years, significant progress has been achieved in the subject of plant nutrition, especially in the formulation and delivery of micronutrients. Improving horticulture crops' nutrient uptake efficiency and effectiveness is the main goal of these developments. The micronutrient application strategy depends on the crop type, soil, requirements, and the planned timing of fertilizer delivery [8]. The most often utilized methods for applying micronutrients are soil application, foliar spraying, hydroponic systems and fertigation [13–16].

Using the foliar application of nutrients helps plants stay healthier and meet their micronutrient needs at critical stages of development [17]. It is well known that foliar fertilizers provide nutrients to the crop's tissues and organs instantly [18]. Moreover, the danger of nutrient fixation, leaching, accumulation, and toxicity in the soil is reduced by the foliar application of micronutrients [19–21]. Despite the importance of these application methods, the optimal approach for chili cultivation in Bangladesh has not been fully explored especially with interaction with variety.

This lack of region-specific knowledge calls for research to determine the most effective methods of micronutrient application for chili, to improve overall yield and fruit quality. Besides micronutrient application, a variety of crops has a great impact on the growth and yield of any crop [22]. This difference occurred due to the variations in the genotypes of varieties. The biochemical compounds also vary from variety to variety ([23] Improved features enable the cultivar to withstand various conditions and increase chili yield [24]. Chili varieties differ in terms of their genetic potential for growth, stress tolerance, and quality attributes, such as fruit size, colour, flavour, and capsaicin content.

Some available studies (Table A) have been conducted worldwide to enhance the growth, yield and nutritional value of *Capsicum annuum*, including the impact of organic manure [25], foliar application of micronutrients [26] and varietal performance [27]. Unfortunately, there is no evidence of the assessment between different micronutrient application techniques, and the interaction of micronutrient application efficiency and varietal performances; all of the studies investigated only the foliar application of micronutrients. In Bangladesh, chili farming frequently suffers from insufficient variety selection and inappropriate micronutrient management practices, resulting in low yields and poor fruit quality [28]. Although much research has looked at the individual impacts of micronutrient treatment on a variety of performance, there is still a significant demand for integrated analysis.

Hence the need for this study to investigate the evaluation of micronutrient application and selection of proper variety for a set of agro-climatic conditions is very important to achieve maximum

potential, because of differential growth and development behaviour due to different genetic characters of varieties. This consequence leads to the urgent call for a systematic and thorough investigation to determine the appropriate cultivars, to recommend economically practical micronutrient application techniques, and their interactive significance between two sequential years to get the highest output. We hypothesized that different micronutrient application techniques in combination with cultivar differences would considerably improve growth, yield and various quality content. Particularly, we expected that simultaneous foliar and soil application would enhance nutrient update capacity and improve both agronomic and nutritional traits compared to solely foliar or soil application. Hence this study aims to evaluate the influence of different micronutrient application methods on the growth, yield, and fruit quality of chili. It seeks to explore how various chili varieties respond to these application techniques while identifying the most effective strategies for enhancing production. Additionally, the research focuses on determining the optimal combination of capsicum variety and micronutrient application method to achieve maximum yield and superior fruit quality.

Table A: The available studies on improving the growth, yield and quality of Chili from worldwide literature.

| Aspect                                                        | Country     | Key findings                                                                                                                                                                                                                                       | References |
|---------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Effect of Varieties and Integrated Nutrient Management        | India       | RDF + Vermicompost significantly improved growth and yield. Control showed lowest growth and yield attributes.                                                                                                                                     | [27]       |
| Bioactive compound and nutraceutical potential                | Mexico      | Capsaicin aids in weight reduction and anti-obesity effects. Chili contains beneficial compounds with health-promoting properties.                                                                                                                 | [29]       |
| Evaluation of chemical profile and antioxidant activity       | Italy       | The study indicates that fresh peppers have the highest phenols, flavonoids, and capsaicinoids content. Boiling and freezing negatively affect phytochemical content and antioxidant activity.                                                     | [30]       |
| Impact of Organic Manure                                      | South Korea | Organic manure enhances chilli pepper growth and nutrient uptake. Organic manures increased growth of root, shoot, and fruit dry weights of chilli pepper by 21.4 %, 52.4 %, and 79.7 %, respectively, at 1 × rate compared to the control values. | [25]       |
| Effect of Mycorrhiza Fungi and Various Mineral Fertilizers    | Poland      | The study reveals that mycorrhizal fungi enhance sweet pepper yield and quality. Reduced fertilizer doses do not significantly affect yield.                                                                                                       | [31]       |
| Growing Environments and Micronutrients Application Influence | India       | Two sprays of zinc (100 ppm) or boron (100 ppm) were found best under greenhouse conditions for increasing yield. Interaction effects of micronutrients to variety were significant for marketable traits of capsicum                              | [32]       |
| The influence of foliar fertilizers                           | Poland      | Nano Active Forte increased sweet pepper                                                                                                                                                                                                           | [33]       |

(continued on next page)

(continued)

| Aspect                                                                         | Country    | Key findings                                                                                                                                                                                                 | References |
|--------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Different NPK Fertilization Levels and Nano-Micronutrients Rates               | Egypt      | fruit yield significantly. Enhanced fruit quality with higher sugars, vitamin C, and carotenoids. NPK and Nano-micronutrients enhance chilli growth and yield. Best results from 100 % NPK and 0.5 g/l Nano. | [34]       |
| Foliar application of micronutrients                                           | India      | Treatment T5 showed best growth and yield traits. Maximum quality parameters found in treatment T5.                                                                                                          | [26]       |
| Foliar Application of Nutrients Solution (Macro and Micro) on Local Germplasms | Bangladesh | The study finds maximum fruit per plant from Kajoli with Solution-1. Highest yield from Akashi with Solution-1 treatment.                                                                                    | [35]       |

## 2. Materials and methods

### 2.1. Plant materials and cultivated conditions

The experiment was carried out in Rajshahi division (located at 24°22'42.5"N latitude and 88°37'57.4"E, 20 m sea level) [36] of Bangladesh during Rabi season to Kharif-1 of two different years. The soil texture was silty loam, and the soil was well drained medium high land [37]. The soil chemical analysis results revealed that the soil was silty clay, with pH 8.12, nitrogen 0.11 %, phosphorus 16.60 mg/L, potassium 0.14 mg/L, sulphur 17.05 mg/L and organic carbon 1.59 % per 100g of soil. The zinc and boron content in the soil was found to lower the values of zinc by 0.039 mg/L and boron by 0.23 mg/L. During the experimental period, the average maximum temperature was 32.6°C and minimum temperature was 22.75°C, the average humidity was 74.75 % and the average rainfall 17.85 mm. Three different types of chili cultivars, namely Picnic (Metal), Sakata-653 and Shikarpuri were used. Fifteen hour-soaked seeds were used for good germination. Seedlings of three chili cultivars of one month's age were transplanted in the experimental field.

### 2.2. Soil analysis

Ten plots were chosen randomly with a depth of 0–15 cm all over the field and soil samples were gathered from these plots before the experiment. All undesired components were removed and air-drying, grinding and sieving through a 2 mm mesh were done, respectively. For facilitating further physical and chemical analysis, the composite sample was kept in a store. Assessment of soil texture, organic matter content, pH and total nitrogen was included in the primary soil analysis process.

Furthermore, exchangeable calcium, potassium and magnesium, in addition to available sulphur, phosphorus, copper, iron, zinc, manganese and boron levels were measured as a part of the soil analysis. The Rajshahi Regional Research Laboratory, Bangladesh (Lab no. 14727) of Soil Resource Development Institute (SRDI) was used for the evaluation of the nutrient profile. A glass electrode pH meter was used to assess the soil pH and the wet oxidation method depicted by Ref. [38] was conducted to determine the organic carbon. The percentage of organic carbon was multiplied by a factor of 1.73 [39] to calculate the organic matter content.

Kjeldahl method was used to assess total nitrogen [40,41]. A 0.5 M NaH<sub>2</sub>CO<sub>3</sub> solution with a pH of 8.5 was used to extract the soil and thus phosphorus availability was assessed, and the concentration was

measured spectrophotometrically at 660 nm after reduction of the phosphomolybdate complex with SnCl<sub>2</sub> [42]. A 1 M N1-L10AC, P117.0 solution were utilized to extract exchangeable magnesium, potassium and calcium and a flame photometer was used to analyze K. Besides, atomic absorption spectrophotometry was used to measure Mg and Ca [40,43]. CaCl<sub>2</sub> at the rate of 0.15 % solution and quantified turbidimetrically at 420 nm were used to extract the available sulphur content [40]. Atomic absorption spectrophotometry was used to determine the availability of iron, zinc, manganese and copper (Krishna and Ranjan, 1991, [44]).

### 2.3. Experimental layout and design

The experiment was organized in a split-plot design with three replications allocating the variety on the main plot, and fertilizer application efficiency to the sub-plot with three replications. The treatments comprise three levels of variety viz. V1=SIKARPURI, V2=SAKATA-653, V3=Picnic-(Metal), three levels of micro-nutrient application efficiency; M1 = broadcasting, M2 = spraying, M3 = broadcasting & spraying, and the interaction of variety and micronutrient application efficiency. Micronutrient was used as 2 kg B ha<sup>-1</sup> + 2 kg Zn ha<sup>-1</sup>. Besides, a basal dose of N-120, P-60, K-80, and S-20 kg ha<sup>-1</sup> was used in the study. The distance between block to block was 60 cm and one plot to plot was 40 cm. The seedlings were planted in the rectangular system of planting at a spacing of 75cm × 75 cm accommodating 12 plants in each plot and the depth of planting was 5.0–8.0 cm underneath the soil surface.

### 2.4. Cultural operation

Cultural operations like ploughing, cross-ploughing, and levelling were conducted in the experimental plot, using a power tiller to prepare it well. A perfectly fine tilth was achieved by clearing all the weeds and breaking the cold of the soil. The plot was again levelled, and water channels were made to facilitate irrigation and drainage. All these preparations were made in the field 30 days before the seedling transplantation. All macro-nutrients: cow dung, urea, triple super phosphate, muriate of potash, and gypsum were applied during the final preparation of land for basal dose at the rate of 5 t/ha, 120 kg/ha, 60 kg/ha, 80 kg/ha, 20 kg/ha respectively. Boric acid and zinc sulphate were the forms in which boron and zinc were used in these experiments. Precautions were taken to prevent the attack of soil-borne pests and diseases by applying Miral 3G with 6.8 kg per hectare at the time of final land preparation.

### 2.5. Plant sampling measurements and data collection

#### 2.5.1. Growth parameters and yield attributes

For facilitation of the data collection, a random selection of 10 plants was made in each treatment. Plant height (cm), leaf number per plant, and branch number per plant were measured and counted at 25, 50, 75, 100 and 125 days of planting. Each measurement was conducted 3 times from 3 different plants. The total number of fruits, damaged fruits per plant, seed number per fruit, fruit length (cm), fruit diameter (cm), and fresh fruit weight (g) were all counted and measured at 15, 30 and 45 DAT.

Calipers were used in the central portion of the randomly selected matured fruits for taking measurements. The dry weight of fruit per plant and seed weight (g) were measured separately in each variety using 100 g of ripe fruits and 100 g of seed weight, respectively. Fresh fruit yield per plant and fresh fruit yield (t/ha) were both measured. Each measurement was taken at least 3 times. For taking the dry weight of the fruits, the ripened fruits were dried under high-temperature sunlight and then moved to an oven at 65 °C for three days.

#### 2.5.2. Chemical analysis

Total chlorophyll, chlorophyll-a, and chlorophyll-b were calculated

from a 100g (fresh weight) sample using the formulae described by Ref. [45]. The phenol-sulphuric acid method, which was depicted by Ref. [46], was utilized to determine the total amount of soluble carbohydrates. Ash content was determined by the methods which were previously described by Ismail 2017. The pH of fresh ripened chili of three different varieties was determined using a pH meter.  $\beta$ -carotene content in fresh ripened chili was determined using an atomic absorption spectrophotometer following the method of [47]. The extraction of  $\beta$ -carotene from the sample was carried out using an acetone-hexane mixture in a 4:6 ratio.

The optical densities of the resulting supernatant were simultaneously measured at wavelengths of 663 nm, 645 nm, 505 nm, and 453 nm using a spectrophotometer. These measurements were then used to determine the  $\beta$ -carotene content. Zinc content in fresh ripened chili was determined using an atomic absorption spectrophotometer depicted by previous researchers Benavides et al., 2021. Calcium content in fresh, mature chili was measured using a colourimetric method as described by Ref. [48]. The vitamin C content was determined using a rapid and non-destructive determination of vitamin C intact red chilies using visible near-infrared spectroscopy and machine learning tools [49]. The total soluble sugar content in ripe chili was analyzed using the Enthroned method [50], while the soluble protein content in ripe chili was determined using the method established by Kong et al., 2015.

## 2.6. Contributing factor and health risk assessment

### 2.6.1. Contributing factor

The following contributing aspects were taken into consideration in order to comprehend the factors that influence *Capsicum annum* yield and nutritional quality: variety, micronutrient application method, and soil properties. These were analyzed using ANOVA to sort out the individual effects of variables and their interaction on plant performance.

### 2.6.2. Health risk assessment

**Daily Intake of Metals (DIM):** The following formula was used to quantitatively assess the average daily intake of metals (DIM) [51–53]:

$$\text{DIM} = \frac{C_{\text{metal}} \times C_{\text{factor}} \times C_{\text{foodintake}}}{\text{BW}_{\text{averagebodyweight}}}$$

Where  $C_{\text{metal}}$  refers to the metal concentration in plants (chili) (mg/kg);  $C_{\text{factor}}$  is the conversion factor (fresh crops into dry weight = 0.085) [51];  $C_{\text{foodintake}}$  refers to the average daily vegetable intake (kg/person/day) and  $\text{BW}_{\text{averagebodyweight}}$  for child and adult 32.7 kg and 73 kg, respectively [51].

**Health Risk Index (HRI):** This index directs the measure of how much daily chili consumption can harm consumers.

$$\text{HRI} = \text{DIM} \div \text{RFD}$$

Where DIM is the Daily Intake of Metals (mg metal/kg body weight/day) as described above and RFD is the reference dose. According to WHO ([54], 1999) and USEPA [55], the RFD values for Cu, Pb, Zn, Ni, Mn, Cr, Cd, and Hg are 0.04, 0.0035, 0.30, 0.02, 0.033, 1.5, 0.001, and 0.003, respectively. When the health risk index (HRI) is less than 1, it is considered safe; when it is greater than 1, it shows negative health effects [56].

### 2.7. Statistical analysis

Statistical package program MSTAT-C was used for data analysis. Duncan's Multiple Range Test (DMRT) was performed to estimate the significance of the difference among the treatments where the level of probability was 5 %. A detailed calculation was completed to determine the means for all the treatments and the F-test was used to perform analyses of variance for most of the characters. The least significant difference test (LSD) was carried out for the evaluation of the

significance of the differences between pairs of means.

## 3. Result

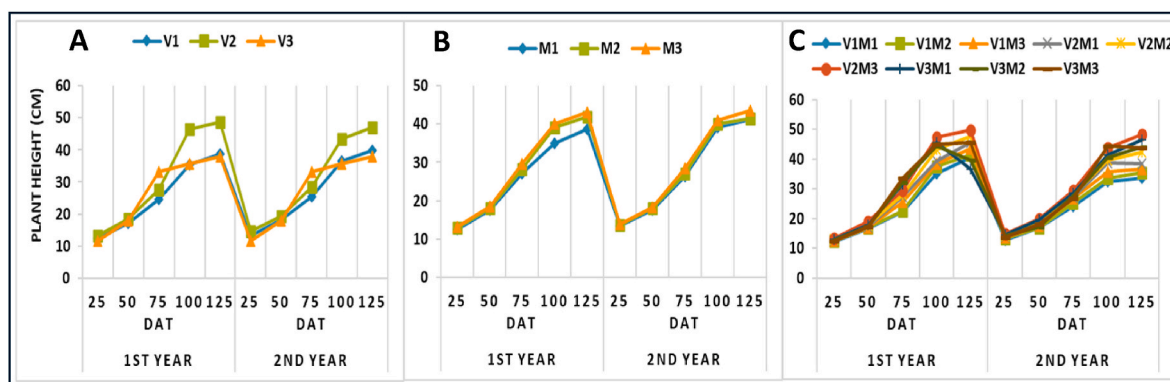
### 3.1. Variety, micronutrient application methods, and their interaction impacts on the growth of chili

Variety, micronutrient application efficiency and their interaction were found significant impact on the treatments. The height of the plants was greatly influenced by variety. The V3 variety displayed the tallest plants (49 cm and 48 cm) in both years at 125 DAT and the V1 variety displayed the smallest plants (40 cm and 38 cm) while V2 showed the intermediate (Fig. 1). The height of the chili was also significantly influenced by micronutrient application efficiency and the interaction of variety and micronutrient application efficiency, with M3 (43 cm and 44 cm) and V2M3 (50 cm and 49 cm) recording the highest plants, respectively in both years after 125 DAT (Fig. 1) where the smallest plant found in the M1 (30 cm & 31 cm) and followed by V2M2 (38 cm & 34 cm) (Fig. 1).

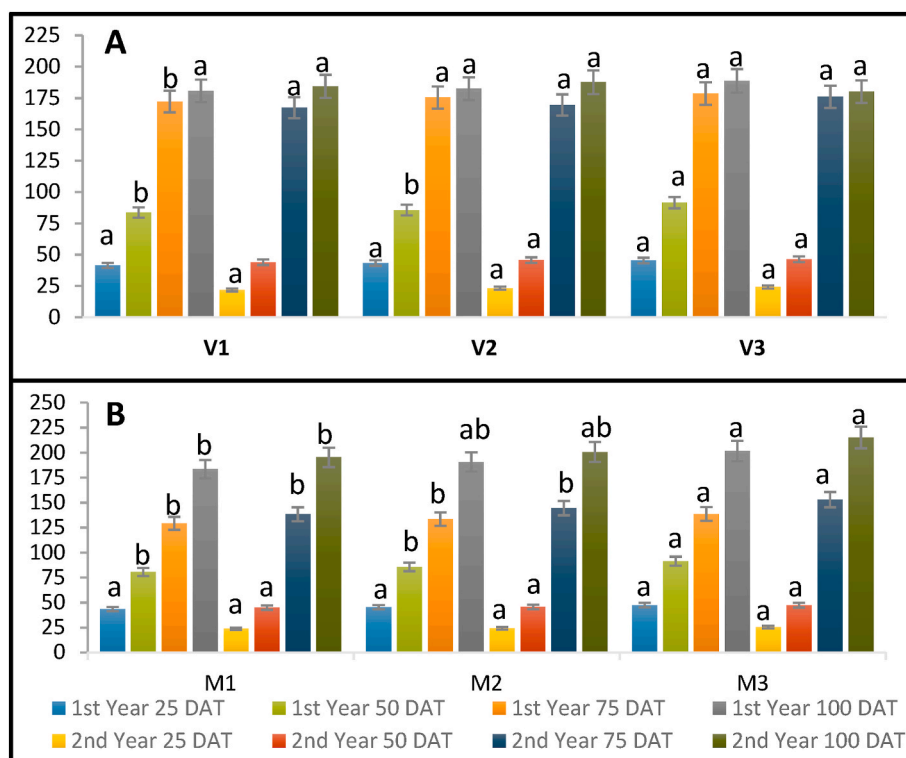
There was no great significance of variety, and micronutrient application efficiency on the number of leaves per chili plant and produced almost similar results. The maximum number of leaves per plant was recorded as 190 in V3 in the first year and 191 in V2 in the second year after 100 DAT (Fig. 2). On the other hand, the micronutrient application efficiency produced maximum leaves in M3 (200 & 220) two years in a row at 100 DAT (Fig. 2). However, the combined effect of variety and micronutrient application efficiency was found on the number of leaves of chili during 100 DAT where the highest output was documented from V3M3 (230, 240) in the first and second years respectively and the lowest number of leaves produced by V2M1 (170, 160) while other interactions found intermediate values (Fig. 3).

On the first flowering days, no discernible effects of the micronutrient application efficiency were seen (Fig. 4). Variety, however, had a significant impact on the initial flowering days; in both years, V2 bloomed late (68 days and 67 days) and the V1 variety showed early flowering at 40 days and 41 days during the first and second year respectively while V3 showed the intermediate value compared to V1 and V2 (Fig. 4). The interaction of variety and micronutrient application efficiency showed great variation in the initiation of flowering. The V1M1 combination started flowering at 39 days in the first year and 40 days in the second year which was statistically similar with V1M2 and V1M3 and the V2M3 demonstrated late flowering (66 days) in the first year and V3M3 in the second year first bloomed at 70 days and showed statistically similar with V2M3 and V3M3 while the interaction of V2M1, V3M1 and V3M2 were showed intermediate value (Fig. 4).

There is a great difference in the number of flowers per plant because of the varieties (Fig. 5). At 30 DAT both the three varieties produced the maximum number of flowers in the first year (33, 31 and 32 respectively) whereas in the 2nd year the maximum number of flowers came at 45 DAT (16, 14 and 12). Therefore, V1 produced the maximum number of flowers in both the 1<sup>st</sup> and 2nd year and V2 brought the minimum number of flowers. Similarly, micronutrient application efficiency showed a maximum number of flowers at 30 and 45 DAT respectively for both the two years. However, the combination of broadcasting and spraying didn't bring any considerable results for both the two sequential years. Spraying micronutrients gave significant results in the 1st year where the highest number of flowers came at 30 DAT (33) and the minimum produced at 15 DAT. In the 2nd year, only M1 exhibited significant findings (Fig. 5). Interaction of variety and micronutrient application efficiency also influenced flower. V2M3 showed considerable findings for both the two years where maximum flowers produced at 30 DAT (32) which is statistically at par with 15 DAT. The lowest flower recorded from the V2M1 and V3M1 were less than 10 (Fig. 6).



**Fig. 1.** Effect of variety (A), micronutrients application efficiency (B), and interaction of variety and micronutrients application efficiency (C) on plant height of chili at different days after transplanting in the consecutive two years experiment.



**Fig. 2.** Effect of variety (A), micronutrients application efficiency (B) on the number of leaves at different days after transplanting in the consecutive two years experiment. Different letters indicate significance difference ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .

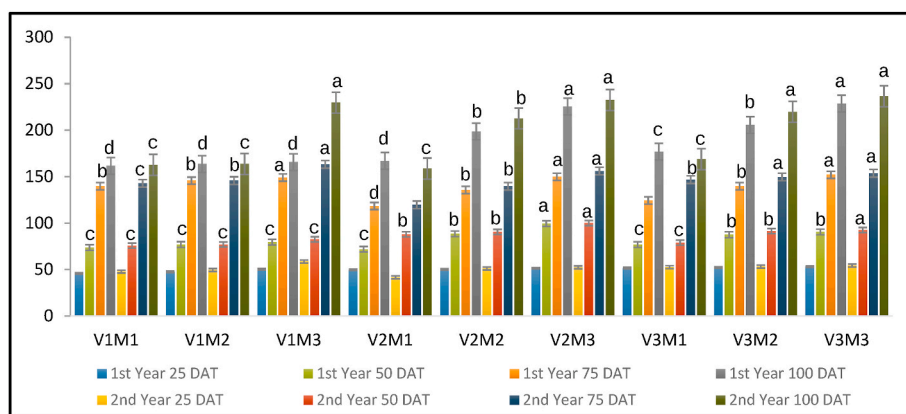
### 3.2. Effect of variety, micronutrients and their interaction on yield and yield attributes

The impact of different varieties showed significant ( $p < .05$ ) differences in fruit length for both the two successive years (Table 1). The maximum fruit length (10.48) was produced by V1 followed by V3, and V2 produced the minimum (7.0) in the first year, this result remained relatively the same with a slight increase of fruit length in the 2nd year from V1 (11.70). However, micronutrients and interaction between variety and micronutrients didn't bring any difference in fruit length (Tables 2 and 3). In general, the maximum fruit length is observed from V1 in the 2nd year.

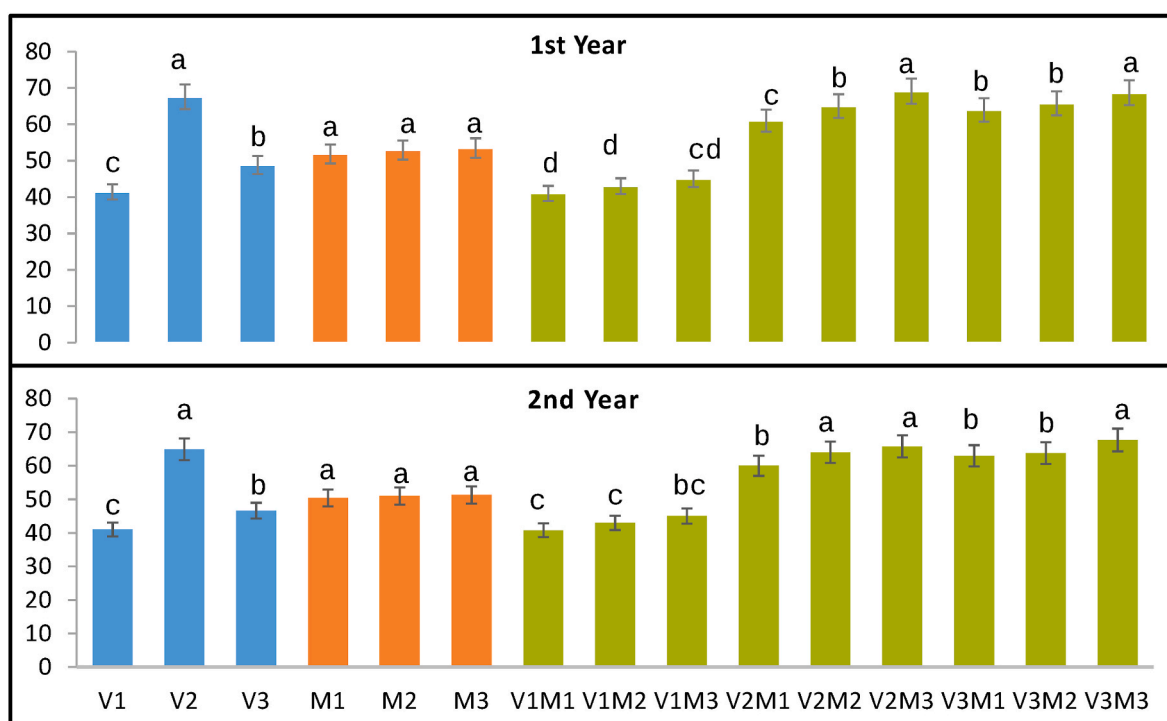
The fresh weight of fruit significantly varied with different varieties for the two sequential years (Table 1). In the second year, the highest weight was found from V3 (6.06) which was statistically at par with V1, whereas the least weight (5.0) was obtained from V2. Although the

application of micronutrients didn't give any change for the fresh weight of fruit in the 1st year, it showed a significant difference in the 2nd year; where M3 produced maximum fresh weight (6.53) and M1 produced minimum (5.03), which was statistically at par with M2 (Table 2). However, the combined application of varieties and micronutrients remained unchanged for fresh fruit weight (Table 3).

The effect of varieties on the dry weight of fruit exhibited a significant difference in the first year where the heaviest fruit was found from V3 (0.55) and the lowest weight obtained from V2 (0.46) (Table 1). However, in the second year, varieties didn't bring any change in the dry weight of fruit. This result was reversed for micronutrient application where the addition of micronutrients remained the same for the dry weight of fruit in the 1st year, however in the 2nd year, M3 produced the heaviest weight (0.68) which was statistically at par with M2 (Table 2). The least dry weight of fruit was found from M1 (0.60). However, the interaction of varieties and micronutrients showed no change in the dry



**Fig. 3.** Effect of the interaction of variety and micronutrients application efficiency on the number of leaves at different days after transplanting in the consecutive two years experiment. Different letters indicate significance difference ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .

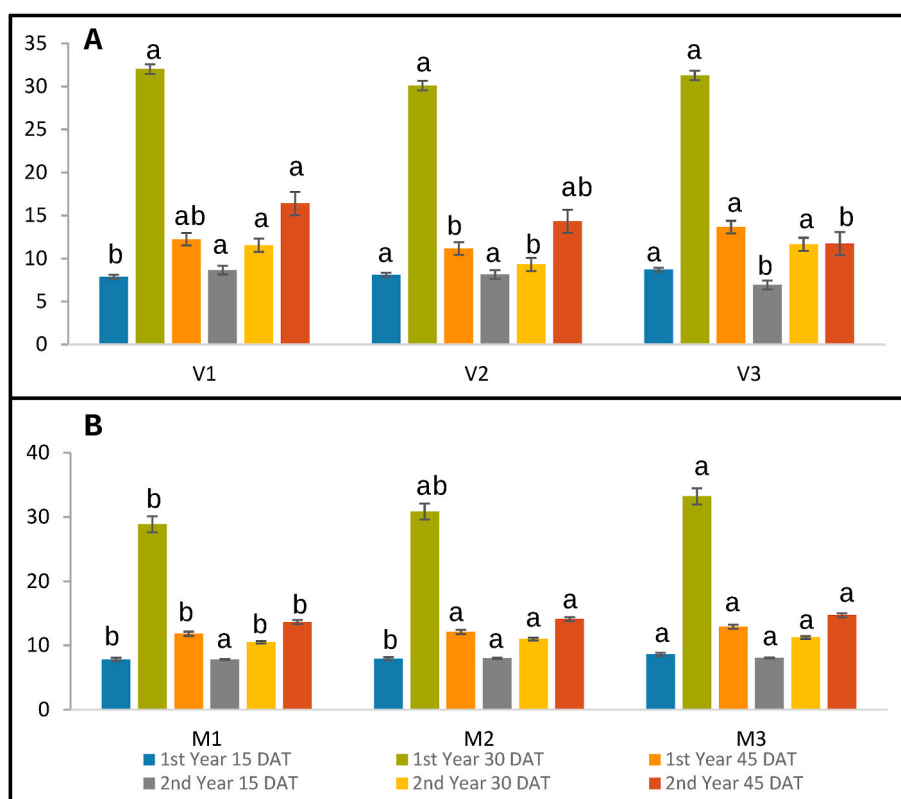


**Fig. 4.** Effect of variety, micronutrients application efficiency and their interaction on the first day of flowering (FAD) in the consecutive two years experiment. Different letters indicate significance difference ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .

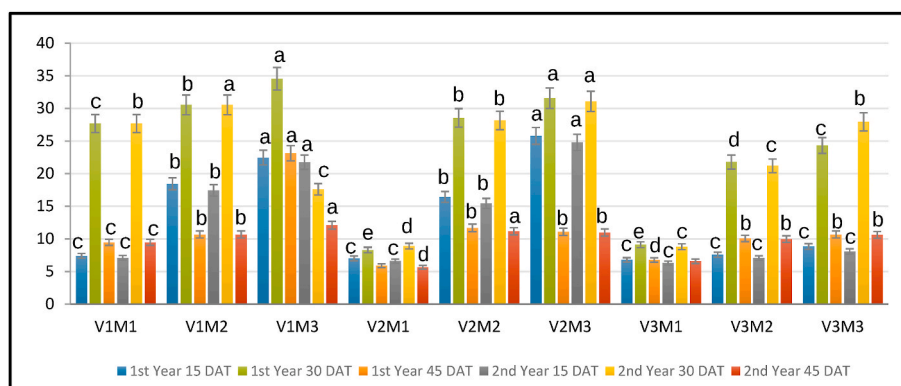
weight of fruit (Table 3).

The seed weight varied considerably with different varieties ( $P < .05$ ) in the first year whereas 2nd year showed identical results (Table 1). The maximum seed weight (0.32) was found in the V3 variety followed by V2 (0.30), while V1 produced the minimum seed weight (0.27). This result revealed that the V3 variety produced the maximum seed weight. Unlike variety, the application of micronutrients brought significant change for two consecutive years (Table 2). Application of M3 gave the heaviest weight (0.36) and the lowest weight (0.232) came from M1 in the 1st year. The interactive effect of varieties and micronutrients didn't show any significant difference in seed weight in the 1st year; however, it produced significant change in the 2nd year where seed weight varied from 0.39 to 0.21. The highest seed weight (0.39) was found by the application of V3M3, while the lowest (0.21) was found in V1M1. Therefore, the combined addition of Picnic (metal) and simultaneous spraying and broadcasting produced the highest seed weight (0.39).

The different varieties varied significantly ( $p < .05$ ) with fresh fruit weight per plant for two successive years (Table 1). V3 produced the highest fresh fruit weight per plant (187.55), followed by V2 which was statistically at par with V1 in the 1st year. However, in the 2nd year, fresh fruit weight reduced than in the first year. Moreover, no significant change occurred with the micronutrient application for fresh fruit weight per plant in the 1st year (Table 2). However, micronutrients gave a significant difference in the 2nd year on fresh fruit weight per plant, although it remained similar to varieties impact in 2nd year (Table 2). Additionally, the interaction between variety and micronutrients depicted significant results for both the two sequential years (Table 3). The maximum fresh weight per plant (176.04) was found from V3M3 in the 1st year, which was significantly different among other treatments. However, the minimum fresh fruit weight per plant was found with the application of V2M1. Notably, in the 2nd year, V3M3 produced more fresh weight (173.37) than 1st year which was significantly different among other interactions and V1M1 produced the minimum weight



**Fig. 5.** Effect of variety (A), micronutrients application efficiency (B) on the number of flowers at different days after transplanting in the consecutive two years experiment. Different letters indicate significance difference ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .



**Fig. 6.** Effect of the interaction of variety and micronutrients application efficiency on the number of flowers at different days after transplanting in the consecutive two years experiment. Different letters indicate significance difference ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .

(157.81) instead of V2M1. Hence, combined applications of V3 and M3 gave the highest fresh weight per plant (193.37) in the next season.

Fresh fruit yield per hectare varied considerably with three varieties for both the two successive years where V3 produced the maximum fresh fruit yield (6.90) followed by V2 (4.90) which was statistically at par with V1 in the second year. In general, the V3 variety gave the highest yield in the 2nd year. Similarly, the application of micronutrients showed a significant variation in fresh fruit yield for both the two consecutive years, and the highest yield (7.52) was obtained from M3 and the lowest from M1 in the 2nd year. Generally, the result revealed that E3 produced the maximum fresh fruit yield in the 2nd year. Like variety and micronutrients, their combined application gave significant differences for both sequential years (Table 3). The highest fresh fruit yield per hectare (9.72) was found with the application of V3M3 which was significantly varied among other interactions and

V1M1 produced the lowest yield (3.18) in the 2nd year. However, the rest of the yield parameters remained unchanged with the variety, micronutrients and interaction between varieties and micronutrients for both the two consecutive years.

### 3.3. Effect of variety, micronutrient application efficiency and their interaction on different quality attributes

The quality parameters of chili fruits were influenced by variety, micronutrient application efficiency and their interaction. The pH levels did not show any noteworthy difference in the 1st year, but the varieties significantly varied in the 2nd year. The highest pH value was observed in the V2 variety (5.650), followed by V1 (5.58), and the lowest value was found in the V3 variety (5.563) (Table 4). However, the impact of micronutrients and the interaction of variety and micronutrients did not

**Table 1**

The growth and yield characteristics of chili under different varieties used in the consecutive two years experiment.

| Treatments      | Damaged fruit plant <sup>-1</sup> | Fruit length (cm) | Fruit diameter (cm) | Fresh weight of fruit (g) | Dry weight of fruit (g) | Seed Number per fruits | Seed Weight (g) | Fresh fruit weight plant <sup>-1</sup> (g) | Fresh fruit Yield (Ton ha <sup>-1</sup> ) |
|-----------------|-----------------------------------|-------------------|---------------------|---------------------------|-------------------------|------------------------|-----------------|--------------------------------------------|-------------------------------------------|
| <b>1st Year</b> |                                   |                   |                     |                           |                         |                        |                 |                                            |                                           |
| V <sub>1</sub>  | 4.343                             | 10.874a           | 0.987               | 5a                        | 0.537b                  | 73.398                 | 0.271b          | 164.684b                                   | 4.6822c                                   |
| V <sub>2</sub>  | 3.688                             | 7.000b            | 0.978               | 4.643b                    | 0.464c                  | 68.392                 | 0.301a          | 167.092b                                   | 4.7638b                                   |
| V <sub>3</sub>  | 3.877                             | 8.492b            | 1.06                | 4.899a                    | 0.557a                  | 80.475                 | 0.324a          | 187.552a                                   | 6.7042a                                   |
| LS              | NS                                | **                | NS                  | **                        | **                      | NS                     | **              | **                                         | **                                        |
| CV              | 9.97                              | 5.57              | 4.67                | 7.59                      | 6.21                    | 6.04                   | 4.82            | 5.19                                       | 6.32                                      |
| <b>2nd Year</b> |                                   |                   |                     |                           |                         |                        |                 |                                            |                                           |
| V <sub>1</sub>  | 3.988                             | 11.709a           | 0.929               | 6.026a                    | 0.669                   | 68.479                 | 0.246           | 148.418 ab                                 | 4.774b                                    |
| V <sub>2</sub>  | 3.658                             | 6.936b            | 0.958               | 5.00b                     | 0.596                   | 69.231                 | 0.264           | 134.981b                                   | 4.9075b                                   |
| V <sub>3</sub>  | 3.768                             | 7.948b            | 1.066               | 6.064a                    | 0.675                   | 78.611                 | 0.285           | 167.472a                                   | 6.9027a                                   |
| LS              | NS                                | **                | NS                  | **                        | NS                      | NS                     | NS              | **                                         | **                                        |
| CV              | 8.01                              | 4.12              | 4.62                | 7.09                      | 3.50                    | 5.51                   | 5.48            | 3.74                                       | 9.17                                      |

In a column, figures bearing similar letter (s) or without letter are identical and different letters indicate significant differences ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD, n = 10 V<sub>1</sub> = SHIKARPURI, V<sub>2</sub> = SAKATA-653, V<sub>3</sub> = PICNIC (Metal).

**Table 2**

Effects of micronutrient application efficiency on the growth and yield parameters of chili in the consecutive two years experiment.

| Parameter       | Damaged fruit plant <sup>-1</sup> | Fruit length (cm) | Fruit diameter (cm) | Fresh weight of fruit (g) | Dry weight of fruit (g) | Seed Number per fruit | Seed Weight (g) | Fresh fruit weight plant <sup>-1</sup> (g) | Fresh fruit Yield (Ton ha <sup>-1</sup> ) |
|-----------------|-----------------------------------|-------------------|---------------------|---------------------------|-------------------------|-----------------------|-----------------|--------------------------------------------|-------------------------------------------|
| <b>1st year</b> |                                   |                   |                     |                           |                         |                       |                 |                                            |                                           |
| M <sub>1</sub>  | 3.775                             | 8.453             | 0.975               | 4.698                     | 0.507                   | 69.186                | 0.232c          | 166.897                                    | 3.397652c                                 |
| M <sub>2</sub>  | 3.986                             | 8.694             | 1.012               | 4.772                     | 0.52                    | 73.61                 | 0.306b          | 172.944                                    | 5.275868b                                 |
| M <sub>3</sub>  | 4.148                             | 9.219             | 1.037               | 5.072                     | 0.531                   | 79.47                 | 0.358a          | 179.486                                    | 7.476806a                                 |
| LS              | NS                                | NS                | NS                  | NS                        | NS                      | NS                    | **              | NS                                         | **                                        |
| CV              | 9.97                              | 5.57              | 4.67                | 7.59                      | 6.21                    | 6.04                  | 4.82            | 5.19                                       | 6.32                                      |
| <b>2nd Year</b> |                                   |                   |                     |                           |                         |                       |                 |                                            |                                           |
| M <sub>1</sub>  | 3.694                             | 8.616             | 0.955               | 5.033b                    | 0.603b                  | 68.772                | 0.252b          | 148.418b                                   | 3.642205c                                 |
| M <sub>2</sub>  | 3.785                             | 8.792             | 0.983               | 5.525b                    | 0.653a                  | 72.403                | 0.267 ab        | 134.981c                                   | 5.417726b                                 |
| M <sub>3</sub>  | 3.935                             | 9.185             | 1.015               | 6.531a                    | 0.684a                  | 75.146                | 0.277a          | 167.472a                                   | 7.524246a                                 |
| LS              | NS                                | NS                | NS                  | **                        | **                      | NS                    | **              | **                                         | **                                        |
| CV              | 8.01                              | 4.12              | 4.62                | 7.09                      | 3.50                    | 5.51                  | 5.48            | 3.74                                       | 9.17                                      |

In a column, figures bearing similar letter (s) or without letter are identical and Different letters indicate significant differences ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD, n = 10, M<sub>1</sub> = Broadcasting, M<sub>2</sub> = Spraying, M<sub>3</sub> =  $\frac{1}{2}$  Broadcasting +  $\frac{1}{2}$  Spraying, Micronutrient doses used in this experiment was 2 kg B ha<sup>-1</sup> + 4 kg Zn ha<sup>-1</sup>.

show any considerable changes in pH (Tables 5 and 6).

The impact of varieties showed substantial ( $p < .05$ ) differences in Vitamin C (Vit-C) contents in both years, with the 1st year showing better results than the 2nd year. The V<sub>3</sub> variety had the highest amount (96.101 mg/100g), followed by V<sub>1</sub>, while the V<sub>2</sub> variety had the lowest value (72.401 mg/100g) (Table 4). The two successive years showed similar results for micronutrients as well as the interaction of variety and micronutrient application methods. In the case of micronutrient application, Vit-C content varied significantly with different fertilizer application methods, where the M<sub>3</sub> ( $\frac{1}{2}$  Broadcasting +  $\frac{1}{2}$  Spraying) method was slightly higher than the M<sub>2</sub> (Spraying) method, and the M<sub>1</sub> (Broadcasting) method had the least (Table 5). Moreover, Vit-C content showed differences, ranging from 70.426 to 96.72 mg/100g, with the V<sub>3</sub>M<sub>3</sub> treatment having the maximum amount and the V<sub>2</sub>M<sub>1</sub> treatment having the minimum amount of Vit-C content (Table 6). The elevated vitamin C (96.101 mg/100g) and sugar content (1.624 %) in V<sub>3</sub>M<sub>3</sub>-treated plants support the findings of Salam et al. [57], which emphasize the critical role of boron and zinc in improving biochemical fruit quality, such as nutrient retention and flavor development.

For variety, the percentage of sugar content improved in the 2nd year compared to the 1st year (Table 4). Both years saw significant differences, with V<sub>3</sub> having the highest percentage (1.624), followed by V<sub>2</sub> (1.616), while V<sub>1</sub> (1.096) had the lowest content. However, the influence of micronutrients and the interaction of variety and micronutrients did not show notable variation (Tables 5 and 6). Carbohydrate content did not show obvious variations in the effect of variety in the 1st

year, but it produced significant differences in the 2nd year, with V<sub>3</sub> (3.938) having the highest percentage, followed by V<sub>2</sub> (3.78), while V<sub>1</sub> (3.163) had the lowest content of carbohydrates (Table 4). Moreover, the amount of carbohydrate content was highest in the M<sub>2</sub> (Spraying) method, followed by M<sub>3</sub> ( $\frac{1}{2}$  Broadcasting +  $\frac{1}{2}$  Spraying), while the lowest content was observed in the M<sub>1</sub> (Broadcasting) method (Table 5). The values for the 2nd year were identical to the 1st year. The percentage of carbohydrates also showed significant differences. The maximum value was in the V<sub>3</sub>M<sub>2</sub> (3.956) treatment, while the minimum percentage was in the V<sub>1</sub>M<sub>1</sub> (2.949) treatment over the two years.

The impact of Chlorophyll and Ca content was significantly influenced by variety and micronutrients. In both cases, the V<sub>3</sub> variety (4.663, 0.92) and M<sub>3</sub> treatment (4.393, 0.83) showed the highest amount respectively, although there was no significant effect from the combination of variety and micronutrients (Tables 4–6). The effect of Zn content showed significant results over the two consecutive years, with V<sub>2</sub> (6.871) having the highest amount, followed by V<sub>1</sub> (5.142), while the V<sub>3</sub> variety (3.602) had the least amount. The 2nd year showed slightly better results compared to the 1st year (Table 4). However, the micronutrient and the interactive combination of variety and micronutrient did not show any notable differences (Tables 5 and 6).

In the 1st year,  $\beta$ -carotene did not show any remarkable changes due to the impact of variety. However, in the 2nd year, significant differences were observed, with the V<sub>1</sub> and V<sub>3</sub> varieties showing the maximum and minimum content respectively (Table 4). It also had a crucial impact on the micronutrient and the integration of variety and

**Table 3**

Effects of the interaction of varieties and micronutrients application efficiency on various yield parameters of chili in the consecutive two years experiment.

| Treatments                    | Damaged fruit plant <sup>-1</sup> | Fruit length (cm) | Fruit diameter (cm) | Fresh weight of fruit (g) | Dry weight of fruit (g) | Seed Number per fruit | Seed Weight (g) | Fresh fruit weight plant <sup>-1</sup> (g) | Fresh fruit Yield (Ton ha <sup>-1</sup> ) |
|-------------------------------|-----------------------------------|-------------------|---------------------|---------------------------|-------------------------|-----------------------|-----------------|--------------------------------------------|-------------------------------------------|
| <b>1st Year</b>               |                                   |                   |                     |                           |                         |                       |                 |                                            |                                           |
| V <sub>1</sub> M <sub>1</sub> | 3.828                             | 11.477            | 0.892               | 5.054                     | 0.623                   | 65.555                | 0.235           | 141.451f                                   | 3.302i                                    |
| V <sub>1</sub> M <sub>2</sub> | 3.944                             | 11.62             | 0.926               | 5.747                     | 0.686                   | 68.945                | 0.245           | 146.108e                                   | 4.472f                                    |
| V <sub>1</sub> M <sub>3</sub> | 4.191                             | 12.03             | 0.968               | 7.276                     | 0.697                   | 70.936                | 0.26            | 157.697d                                   | 6.549c                                    |
| V <sub>2</sub> M <sub>1</sub> | 3.551                             | 6.708             | 0.93                | 4.674                     | 0.554                   | 65.87                 | 0.249           | 128.234i                                   | 3.353h                                    |
| V <sub>2</sub> M <sub>2</sub> | 3.655                             | 6.831             | 0.961               | 4.814                     | 0.602                   | 69.556                | 0.264           | 135.594h                                   | 5.013e                                    |
| V <sub>2</sub> M <sub>3</sub> | 3.768                             | 7.27              | 0.983               | 5.512                     | 0.632                   | 72.268                | 0.278           | 141.116g                                   | 6.353d                                    |
| V <sub>3</sub> M <sub>1</sub> | 3.703                             | 7.662             | 1.043               | 5.372                     | 0.632                   | 74.891                | 0.271           | 162.154c                                   | 4.271g                                    |
| V <sub>3</sub> M <sub>2</sub> | 3.756                             | 7.927             | 1.062               | 6.014                     | 0.67                    | 78.709                | 0.291           | 164.221b                                   | 6.768b                                    |
| V <sub>3</sub> M <sub>3</sub> | 3.847                             | 13.256            | 1.093               | 6.805                     | 0.722                   | 82.232                | 0.294           | 176.041a                                   | 9.671a                                    |
| LS                            | NS                                | NS                | NS                  | NS                        | NS                      | NS                    | NS              | **                                         | **                                        |
| CV                            | 8.01                              | 4.12              | 4.62                | 7.09                      | 3.50                    | 5.51                  | 5.48            | 3.74                                       | 9.17                                      |
| <b>2nd Year</b>               |                                   |                   |                     |                           |                         |                       |                 |                                            |                                           |
| V <sub>1</sub> M <sub>1</sub> | 4.126                             | 10.473            | 0.959               | 4.756                     | 0.523                   | 66.854                | 0.218d          | 157.818h                                   | 3.182611f                                 |
| V <sub>1</sub> M <sub>2</sub> | 4.419                             | 10.683            | 0.985               | 4.811                     | 0.541                   | 72.073                | 0.274cd         | 165.134                                    | 4.409551d                                 |
| V <sub>1</sub> M <sub>3</sub> | 4.486                             | 11.467            | 1.015               | 5.433                     | 0.549                   | 81.268                | 0.321bc         | 171.1e                                     | 6.454516b                                 |
| V <sub>2</sub> M <sub>1</sub> | 3.509                             | 6.81              | 0.924               | 4.521                     | 0.451                   | 64.249                | 0.225d          | 160.62g                                    | 3.197189f                                 |
| V <sub>2</sub> M <sub>2</sub> | 3.679                             | 6.945             | 0.995               | 4.612                     | 0.465                   | 69.004                | 0.316bc         | 166.667f                                   | 4.839333c                                 |
| V <sub>2</sub> M <sub>3</sub> | 3.876                             | 7.247             | 1.014               | 4.797                     | 0.478                   | 71.924                | 0.361 ab        | 173.988d                                   | 6.254934b                                 |
| V <sub>3</sub> M <sub>1</sub> | 3.689                             | 8.077             | 1.041               | 4.818                     | 0.548                   | 76.455                | 0.252d          | 182.253c                                   | 3.813155e                                 |
| V <sub>3</sub> M <sub>2</sub> | 3.861                             | 8.455             | 1.056               | 4.892                     | 0.556                   | 79.752                | 0.329bc         | 187.031b                                   | 6.578719b                                 |
| V <sub>3</sub> M <sub>3</sub> | 4.081                             | 12.944            | 1.083               | 4.987                     | 0.567                   | 85.219                | 0.392a          | 193.371a                                   | 9.720969 a                                |
| LS                            | NS                                | NS                | NS                  | NS                        | NS                      | NS                    | **              | **                                         | **                                        |
| CV                            | 9.97                              | 5.57              | 4.67                | 7.59                      | 6.21                    | 6.04                  | 4.82            | 5.19                                       | 6.32                                      |

In a column, figures bearing similar letter (s) or without letter are identical and Different letters indicate significant differences ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .

V<sub>1</sub> = SHIKARPURI, V<sub>2</sub> = SAKATA-653, V<sub>3</sub> = PICNIC (Metal).

M<sub>1</sub> = Broadcasting, M<sub>2</sub> = Spraying, M<sub>3</sub> =  $\frac{1}{2}$  Broadcasting +  $\frac{1}{2}$  Spraying, Micronutrient doses used in this experiment was 2 kg B ha<sup>-1</sup> + 4 kg Zn ha<sup>-1</sup>.

**Table 4**

Effects of the varieties on the quality parameters of chili in the consecutive two years experiment.

| Parameters      | pH     | Vit-C'(mg/100g) | Sugar (%) | Carbohydrate (%) | Chlorophyll (mg/100g) | Ca(mg/F) | Zn(mg/F) | $\beta$ -Carotene( $\mu$ g/100g) | Ash (%) | Protine (%) |
|-----------------|--------|-----------------|-----------|------------------|-----------------------|----------|----------|----------------------------------|---------|-------------|
| <b>1st Year</b> |        |                 |           |                  |                       |          |          |                                  |         |             |
| V <sub>1</sub>  | 5.407  | 84.702b         | 1.071b    | 3.305            | 4.544a                | 0.845a   | 5.142b   | 3.611                            | 1.501   | 0.908a      |
| V <sub>2</sub>  | 5.631  | 74.642c         | 1.559a    | 3.848            | 3.853b                | 0.658b   | 6.869a   | 2.912                            | 1.449   | 0.775b      |
| V <sub>3</sub>  | 5.667  | 96.101a         | 1.582a    | 4.025            | 4.663a                | 0.924a   | 3.602c   | 3.108                            | 1.525   | 1.014a      |
| LS              | NS     | **              | **        | NS               | **                    | **       | **       | NS                               | NS      | **          |
| CV              | 1.1    | 2.40            | 2.89      | 2.58             | 1.66                  | 2.72     | 1.93     | 1.92                             | 1.28    | 4.24        |
| <b>2nd Year</b> |        |                 |           |                  |                       |          |          |                                  |         |             |
| V <sub>1</sub>  | 5.586b | 84.211b         | 1.096b    | 3.163b           | 4.723a                | 0.861b   | 5.341b   | 3.696a                           | 1.494b  | 0.913b      |
| V <sub>2</sub>  | 5.658a | 72.401c         | 1.616a    | 3.78a            | 3.743b                | 0.666c   | 6.871a   | 2.778b                           | 1.544a  | 0.776c      |
| V <sub>3</sub>  | 5.563c | 94.316a         | 1.624a    | 3.938a           | 4.835a                | 0.91a    | 3.644c   | 2.869b                           | 1.569a  | 0.994a      |
| LS              | **     | **              | **        | **               | **                    | **       | **       | **                               | **      | **          |
| CV              | 1.24   | 3.63            | 3.11      | 3.95             | 2.99                  | 3.96     | 2.55     | 3.71                             | 1.7     | 2.98        |

In a column, figures bearing similar letter (s) or without letter are identical and Different letters indicate significant differences ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ , V<sub>1</sub> = SHIKARPURI, V<sub>2</sub> = SAKATA-653, V<sub>3</sub> = PICNIC (Metal).

micronutrient, with both years showing similar results. In the case of micronutrients,  $\beta$ -carotene exhibited the largest amount in the M<sub>3</sub> method (3.168), followed by M<sub>1</sub> (3.109), and the M<sub>2</sub> method had the least amount (3.065) (Table 5). However, the highest value was found in the V<sub>1</sub>M<sub>1</sub> interaction method (3.718), and the lowest amount was shown in the V<sub>2</sub>M<sub>2</sub> treatment (2.747) (Table 6). The Ash percentage had no significant effects on varieties in the 1st year, but it showed notable differences in the 2nd year. The maximum ash percentage (1.569) was found in the V<sub>3</sub> variety, followed by V<sub>2</sub> (1.544), while V<sub>1</sub> produced the minimum amount (1.494) (Table 4). However, it did not influence the micronutrient application method (Table 5). On the other hand, the combined effect of variety and micronutrients had a crucial impact on assets, where both years showed similar results. The maximum amount of ash was 1.58 % in the V<sub>3</sub>M<sub>1</sub> interaction method, and the minimum

was 1.486 % in the V<sub>1</sub>M<sub>1</sub> method (Table 6).

The percentage of protein showed a substantial influence on varieties, micronutrients, and the interaction of variety and micronutrients over the two consecutive years. In the 1st year, the V<sub>3</sub> variety had the maximum content followed by V<sub>1</sub>, while V<sub>2</sub> had the lowest content, showing a negligible difference compared to the 2nd year (Table 4). The two consecutive years produced exactly similar results for micronutrients and the combination of variety and micronutrient methods, with M<sub>3</sub> (0.94) having the highest amount, followed by M<sub>2</sub> (0.883), and M<sub>1</sub> having the least percentage (0.859) (Table 5). The highest amount was observed in the V<sub>3</sub>M<sub>3</sub> treatment (1.085), while the lowest amount was shown in the V<sub>2</sub>M<sub>1</sub>.

**Table 5**

Effects of micronutrient application efficiency on the different quality parameters of chili in the consecutive two years experiment.

| Parameters      | pH    | Vit-C'(mg/<br>100g) | Sugar<br>(%) | Carbohydrate (%) | Chlorophyll (mg/<br>100g) | Ca(mg/<br>F) | Zn(mg/<br>F) | β-Carotene(μg/<br>100g) | Ash<br>(%) | Protine<br>(%) |
|-----------------|-------|---------------------|--------------|------------------|---------------------------|--------------|--------------|-------------------------|------------|----------------|
| <b>1st Year</b> |       |                     |              |                  |                           |              |              |                         |            |                |
| M <sub>1</sub>  | 5.582 | 81.649b             | 1.43         | 3.558b           | 4.431a                    | 0.789b       | 5.233        | 3.109 ab                | 1.536      | 0.859c         |
| M <sub>2</sub>  | 5.599 | 84.345 ab           | 1.438        | 3.703a           | 4.476a                    | 0.817 ab     | 5.287        | 3.065b                  | 1.529      | 0.883b         |
| M <sub>3</sub>  | 5.626 | 84.934a             | 1.469        | 3.619 ab         | 4.393b                    | 0.83a        | 5.336        | 3.168a                  | 1.542      | 0.94a          |
| LS              | NS    | **                  | NS           | **               | **                        | **           | NS           | **                      | NS         | **             |
| CV              | 1.24  | 3.63                | 3.11         | 3.95             | 2.99                      | 3.96         | 2.55         | 3.71                    | 1.7        | 2.98           |
| <b>2nd Year</b> |       |                     |              |                  |                           |              |              |                         |            |                |
| M <sub>1</sub>  | 5.582 | 81.649b             | 1.43         | 3.558b           | 4.431a                    | 0.789b       | 5.233        | 3.109 ab                | 1.536      | 0.859c         |
| M <sub>2</sub>  | 5.599 | 84.345 ab           | 1.438        | 3.703a           | 4.476a                    | 0.817 ab     | 5.287        | 3.065b                  | 1.529      | 0.883b         |
| M <sub>3</sub>  | 5.626 | 84.934a             | 1.469        | 3.619 ab         | 4.393b                    | 0.83a        | 5.336        | 3.168a                  | 1.542      | 0.94a          |
| LS              | NS    | **                  | NS           | **               | **                        | **           | NS           | **                      | NS         | **             |
| CV              | 1.24  | 3.63                | 3.11         | 3.95             | 2.99                      | 3.96         | 2.55         | 3.71                    | 1.7        | 2.98           |

In a column, figures bearing similar letter (s) or without letter are identical and Different letters indicate significant differences ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ , M<sub>1</sub> = Broadcasting, M<sub>2</sub> = Spraying, M<sub>3</sub> =  $\frac{1}{2}$  Broadcasting +  $\frac{1}{2}$  Spraying, Micronutrient doses used in this experiment was 2 kg B ha<sup>-1</sup> + 4 kg Zn ha<sup>-1</sup>.

**Table 6**

Effects of variety and fertilizer application efficiency on the different quality parameters of chili in the consecutive two years experiment.

| Parameters                    | pH    | Vit-C'(mg/<br>100g) | Sugar<br>(%) | Carbohydrate<br>(%) | Chlorophyll (mg/<br>100g) | Ca(mg/<br>F) | Zn(mg/<br>F) | β-Carotene(μg/<br>100g) | Ash(%)  | Protine<br>(%) |
|-------------------------------|-------|---------------------|--------------|---------------------|---------------------------|--------------|--------------|-------------------------|---------|----------------|
| <b>1st Year</b>               |       |                     |              |                     |                           |              |              |                         |         |                |
| V <sub>1</sub> M <sub>1</sub> | 5.543 | 83.272ef            | 1.099        | 2.949d              | 4.762                     | 0.848        | 5.317        | 3.718a                  | 1.486g  | 0.894c         |
| V <sub>1</sub> M <sub>2</sub> | 5.55  | 85.931d             | 1.086        | 3.303c              | 4.754                     | 0.861        | 5.329        | 3.677a                  | 1.499e  | 0.886c         |
| V <sub>1</sub> M <sub>3</sub> | 5.664 | 83.43e              | 1.103        | 3.237c              | 4.653                     | 0.874        | 5.377        | 3.693a                  | 1.497f  | 0.961b         |
| V <sub>2</sub> M <sub>1</sub> | 5.643 | 70.426i             | 1.603        | 3.778 ab            | 3.722                     | 0.649        | 6.785        | 2.787c                  | 1.542c  | 0.773d         |
| V <sub>2</sub> M <sub>2</sub> | 5.67  | 72.125h             | 1.612        | 3.852 ab            | 3.795                     | 0.67         | 6.909        | 2.747c                  | 1.529cd | 0.781d         |
| V <sub>2</sub> M <sub>3</sub> | 5.662 | 74.652g             | 1.634        | 3.709b              | 3.711                     | 0.678        | 6.92         | 2.8c                    | 1.561b  | 0.774d         |
| V <sub>3</sub> M <sub>1</sub> | 5.559 | 91.249c             | 1.588        | 3.946a              | 4.808                     | 0.87         | 3.598        | 2.823c                  | 1.581a  | 0.912c         |
| V <sub>3</sub> M <sub>2</sub> | 5.576 | 94.979b             | 1.615        | 3.956a              | 4.88                      | 0.921        | 3.621        | 2.772c                  | 1.559b  | 0.984b         |
| V <sub>3</sub> M <sub>3</sub> | 5.552 | 96.72a              | 1.669        | 3.912 ab            | 4.815                     | 0.939        | 3.712        | 3.01b                   | 1.567a  | 1.085a         |
| LS                            | NS    | **                  | NS           | **                  | NS                        | NS           | NS           | **                      | **      | **             |
| CV                            | 1.24  | 3.63                | 3.11         | 3.95                | 2.99                      | 3.96         | 2.55         | 3.71                    | 1.7     | 2.98           |
| <b>2nd Year</b>               |       |                     |              |                     |                           |              |              |                         |         |                |
| V <sub>1</sub> M <sub>1</sub> | 5.543 | 83.272ef            | 1.099        | 2.949d              | 4.762                     | 0.848        | 5.317        | 3.718a                  | 1.486g  | 0.894c         |
| V <sub>1</sub> M <sub>2</sub> | 5.55  | 85.931d             | 1.086        | 3.303c              | 4.754                     | 0.861        | 5.329        | 3.677a                  | 1.499e  | 0.886c         |
| V <sub>1</sub> M <sub>3</sub> | 5.664 | 83.43e              | 1.103        | 3.237c              | 4.653                     | 0.874        | 5.377        | 3.693a                  | 1.497f  | 0.961b         |
| V <sub>2</sub> M <sub>1</sub> | 5.643 | 70.426i             | 1.603        | 3.778 ab            | 3.722                     | 0.649        | 6.785        | 2.787c                  | 1.542c  | 0.773d         |
| V <sub>2</sub> M <sub>2</sub> | 5.67  | 72.125h             | 1.612        | 3.852 ab            | 3.795                     | 0.67         | 6.909        | 2.747c                  | 1.529cd | 0.781d         |
| V <sub>2</sub> M <sub>3</sub> | 5.662 | 74.652g             | 1.634        | 3.709b              | 3.711                     | 0.678        | 6.92         | 2.8c                    | 1.561b  | 0.774d         |
| V <sub>3</sub> M <sub>1</sub> | 5.559 | 91.249c             | 1.588        | 3.946a              | 4.808                     | 0.87         | 3.598        | 2.823c                  | 1.581a  | 0.912c         |
| V <sub>3</sub> M <sub>2</sub> | 5.576 | 94.979b             | 1.615        | 3.956a              | 4.88                      | 0.921        | 3.621        | 2.772c                  | 1.559b  | 0.984b         |
| V <sub>3</sub> M <sub>3</sub> | 5.552 | 96.72a              | 1.669        | 3.912 ab            | 4.815                     | 0.939        | 3.712        | 3.01b                   | 1.567a  | 1.085a         |
| LS                            | NS    | **                  | NS           | **                  | NS                        | NS           | NS           | **                      | **      | **             |
| CV                            | 1.24  | 3.63                | 3.11         | 3.95                | 2.99                      | 3.96         | 2.55         | 3.71                    | 1.7     | 2.98           |

In a column, figures bearing similar letter (s) or without letter are identical and Different letters indicate significant differences ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .

V<sub>1</sub> = SHIKARPURI, V<sub>2</sub> = SAKATA-653, V<sub>3</sub> = PICNIC (Metal).

M<sub>1</sub> = Broadcasting, M<sub>2</sub> = Spraying, M<sub>3</sub> =  $\frac{1}{2}$  Broadcasting +  $\frac{1}{2}$  Spraying, Micronutrient doses used in this experiment was 2 kg B ha<sup>-1</sup> + 4 kg Zn ha<sup>-1</sup>.

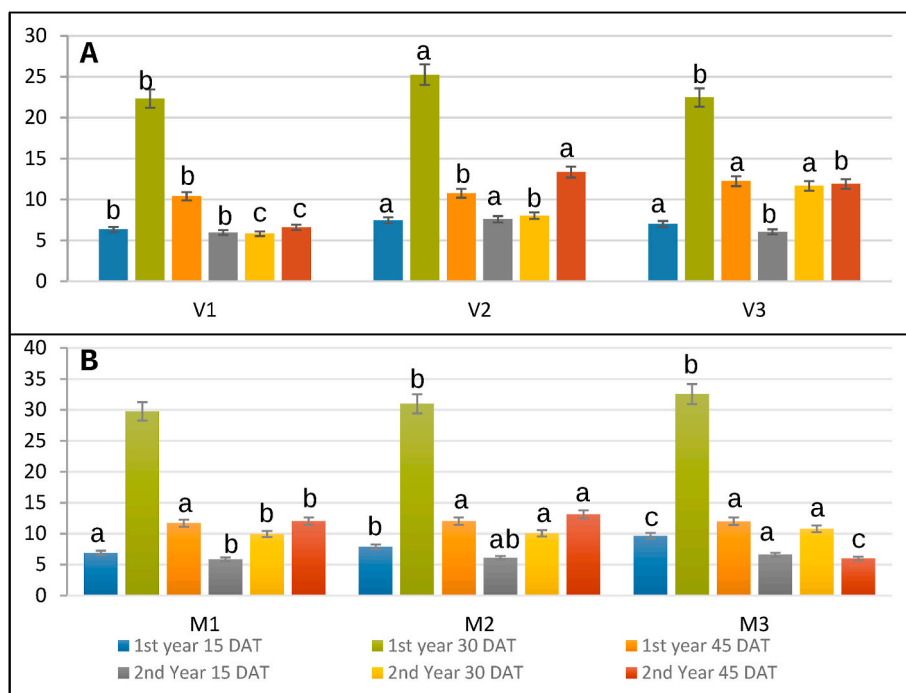
## 4. Discussion

### 4.1. Effect of variety, micronutrient application efficiency and their interaction on growth and yield attributes

Different chili cultivars exhibited different responses to specific nutrients for their proper growth and development as well as yield and quality [58]. Studying the effect of variety, micronutrient application efficiency, and their interaction on growth and yield attributes helps identify the most productive crop varieties under specific nutrient regimes. It provides deep insights into optimal fertilization strategies to enhance plant growth and yield [59]. The proper physiological process assures the maximum yield [60]. Our results revealed that variety has a tremendous impact on growth quantitative traits such as plant height

(Fig. 1A), number of leaves (Fig. 2A), first day of flowering (Fig. 4), number of flowers (Fig. 5A), and number of fruits (Fig. 7A). These development characteristics are commonly interpreted to assess phenotypic and genotypic diversity among varieties under suitable selection (Sreelathakumary et al., 2004, Hossain et al., 2021) and its interaction with environmental condition [22]. Hence, the acquired data suggested that an inherited genotypic feature causes a variation in plant size and other growth parameters among the chili varieties in the investigated two sequential years, where the V<sub>3</sub> or Picnic (metal) variety demonstrated excellent performance.

Not only growth but also yield parameters (fruit length, fruit diameter, fresh weight of fruit, dry weight of fruit, seed weight, fresh fruit yield, etc.) are profoundly controlled by genetic diversity among different traits of chili (Dahal et al., 2015) and wheat (Hossain et al.,



**Fig. 7.** Effect of variety (A), micronutrients application efficiency (B) on the number of fruits per plants at different days after transplanting in the consecutive two years experiment. Different letters indicate significance difference ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .

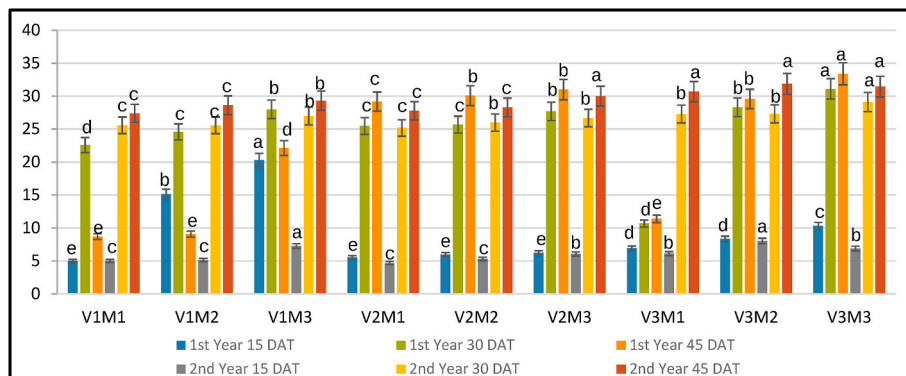
2021). The yield parameters of the two consecutive years (Table 1) demonstrated a corroboration with their findings and the V<sub>3</sub> variety showed maximum response to the parameters which can be illustrated by the variety effect. However, plants require a trace amounts of micronutrients to drive different vegetative growth functions (Azad et al., 2021), and to attain optimal yield and quality parameters to various extent [61].

Micronutrients are most commonly used as single or mixed with other micronutrients and can perform better in vegetative growth and yield of plants (Supti et al. 2022). Different application methods with different doses influence micronutrient efficiency [62,63]. Our findings suggested that the efficiency of the combined broadcasting and spraying application method of mixed (B and Zn) micronutrients enhanced vegetative growth. This is because foliar application ensures quick absorption of nutrients and soil application solves long-term nutrient deficiency [8]. Hence, plant height (Fig. 1B), number of leaves (Fig. 2B), first day of flowering (Fig. 4), number of flowers (Fig. 5B), and number of fruits (Fig. 7B) and yield parameters such as fresh weight of fruit, dry weight of fruit, seed weight, fresh fruit weight are effectively influenced

(Table 2) by combined broadcasting and spraying application of micronutrients.

It is noted that M<sub>3</sub> showed maximum effect on the number of leaves in the 2nd year at 100 DAT (Fig. 2B) and maximum fruit number per plant as well (Table 2). These findings are consistent with previous research (Yadav et al., 2023). Moreover, it is seen that micronutrients have a great impact on growth and yield factors than variety influence (Figs. 1, 2, 4, 5 and 7) in both the research years. According to the outcomes of our research, the interaction of variety and micronutrient application efficiency not only enhances the height of the plant (Fig. 1C) in the two chronological years for almost all the combinations of Variety and Micronutrients but also increased the number of leaves (Fig. 3), early first day of flowering (Fig. 4), number of flowers (Fig. 6), and number of fruits (Fig. 8).

It is noticeable that the number of leaves in response to V1M3, V2M3, and V3M3 combinations was greatly affected in the 2nd year of the experiment at 100 DAT. This is because both variety and micronutrient application efficiency simultaneously work together, and their beneficial effect contributes to reach maximum growth parameters [64,65] in



**Fig. 8.** Effect of the interaction of variety and micronutrients application efficiency on the number of fruits of chili at different days after transplanting in the consecutive two years experiment. Different letters indicate significance difference ( $p < .05$ ; Turkey's HSD test). Values are the mean  $\pm$  SD,  $n = 10$ .

all varieties. Additionally, it has been observed that growth parameters obtained by the interaction effect (Figs. 1C, 3 and 4, 6 & 8) are larger than single effect like variety (Figs. 1A, 2A and 4, 4, 5A & 7A) or micronutrients application efficiency (Figs. 1B, 2B and 4, 5B & 7B). Moreover, yield parameters such as seed weight, fresh fruit weight per plant, and fresh fruit yield have been raised to the peak by the combined effect of variety and micronutrient application efficiency in the two consecutive years. It has been mentioned in the result section that the V<sub>3</sub>M<sub>3</sub> produced the highest number of seeds, fresh fruit weight per plant, and fresh fruit yield which indicate the superiority of interaction effect on V<sub>3</sub> or Picnic (metal) variety of chili. These results are similar to the findings of [66] where it was explained that the yield influence is caused by variety and micronutrient application efficiency mode. The interaction between variety and micronutrient application showed a synergistic effect, further boosting plant height, number of leaves, flowering, fruit set, and yield parameters beyond the individual effects. The V<sub>3</sub>M<sub>3</sub> combination emerged as the most effective, suggesting that strategic integration of high-performing varieties with efficient nutrient management practices is essential for optimizing chili production.

#### 4.2. Effect of variety, micronutrient application efficiency and their interaction on quality parameters

Micronutrient application techniques greatly affect the growth and quality attributes of horticultural crops [8]. Fresh chili's vitamin-C content was significantly influenced by variety (Table 4) which is due to the improvement in ascorbic acid content through fertigation (Ramachandrapa et al., 2010; Azad et al., 2021). This study's outcomes demonstrated that the effect of variation in both years causes a considerable difference in the sugar content of chili. When compared to other varieties during the two-year experiment, the Picnic variety (V<sub>3</sub>) produced the highest amount of carbohydrates. Moreover, Variety has a great effect on the rise in chlorophyll content in chili which is similar to a previous discovery [67]. Genetic diversity causes a variation in chlorophyll and beta-carotene content in the 2nd year shown in Table 4. Micronutrient application method M<sub>3</sub> (½ Broadcasting + ½ Spraying) elicited the best performance of the V<sub>3</sub> variety in getting maximum Vitamin C and beta carotene content according to the results (Table 5).

The application of micronutrients by the integrated methods of broadcasting and spraying increased the efficiency of nutrient uptake due to balanced timing because it has been reported that foliar application takes less time for nutrient absorption while broadcasting takes more time [66]. Moreover, this integrated approach enhanced the various quality content including Vit-C, Ca content, β-carotene and protein. This findings is in harmonic with the findings of [68]. and Chlorophyll content in chili is enhanced to the highest (Table 5) in reaction to micronutrients application method M<sub>2</sub> (Spraying) which has occurred by the combined effect of Zn and boron with spraying application mode [57]. From the results, noteworthy increment is observed in Vitamin C content, Carbohydrate content, Ash percentage and protein content from the interaction of variety and micronutrient application mode.

Previous studies enumerated that the combined application of Zn and B can increase Vitamin C content [57]. The combination effect also showed a significant improvement in ash percentage, which is closely aligned with [69] where it was found that the interaction of variety and micronutrients (Zn and B) can enhance ash content by foliar application mode. Protein content can be increased substantially if the Zn and B combination is applied in an appropriately selected variety [65]. The exceptional performance of the Picnic variety (V<sub>3</sub>) can be attributed to its genetic traits, which enhance nutrient absorption and stress resilience, as observed in our two-year study. When paired with the combined micronutrient method (M<sub>3</sub>)—using both soil broadcasting and foliar spraying, this variety achieved optimal nutrient availability. This dual approach ensured sustained zinc and boron uptake, which boosted enzymatic activity and photosynthesis, leading to higher fresh fruit

weight (193.37 g/plant) and vitamin C content (96.1 mg/100g).

#### 4.3. Case study: real-life application and industrial relevance

To highlight the engineering and commercial relevance of the findings our study's framework and methodology can be directly translated into field-level applications by agri-tech firms, agrochemical companies, and large-scale chili producers. For instance, companies involved in precision agriculture can utilize the data-driven micronutrient application strategies (broadcasting, spraying, and combined) optimized in this study to enhance crop yield and nutritional quality, while minimizing input costs. The model is easily adaptable in practice, as it relies on standard agronomic equipment and procedures, such as hand-held sprayers, soil nutrient profiling, and regular monitoring of plant growth parameters.

We have included a real-world case from a farming cooperative in Rajshahi that applied our optimized fertilization protocol over two cropping seasons. The cooperative reported up to a 22 % increase in fruit yield, improved fruit nutritional value, and a reduction in fertilizer waste. This underscores the practical benefit of our research when scaled beyond experimental settings. Additionally, our health risk assessments confirmed that micronutrient levels in chili fruits (e.g., Zn, B) adhered to WHO safety thresholds. This demonstrates that the V<sub>3</sub>M<sub>3</sub> strategy not only enhances productivity but also ensures food safety—a critical consideration for sustainable farming practices. Farmers can adopt this method confidently, knowing it balances yield gains with consumer health protections.

## 5. Conclusions

The study demonstrated the evaluation of micronutrient application methods and varietal selection in order to get the highest growth, yield and nutritional value of *Capsicum annum*.

The findings from this study suggest that the Picnic variety (V<sub>3</sub>) outperformed V<sub>1</sub> and V<sub>2</sub> in terms of plant height, leaf numbers, fruit yield and nutritional content. Likewise, micronutrient application techniques (M<sub>3</sub>) resulted in the best outcomes in various attributes including Vit-C, protein and chlorophyll content. However, the integrated application of V<sub>3</sub>M<sub>3</sub> was the most effective strategy for the highest yield per hectare and improved quality features including protein, sugar and β-carotene levels.

This study was conducted in a single agroecological zone, limiting the result from getting wide range of geographical regions and considered only two consecutive season which didn't capture long-term climate variability and cost-benefit implication for the different micronutrient application methods.

The result of this study clearly established the optimal variety selection and integrated micronutrient application methods. However, to achieve more precise and sustainable outcomes and build more confidence among farmers; it is suggested that future studies should be conducted considering long-term environmental variation in a wide range of location with economic analysis.

#### CRedit authorship contribution statement

**Md Mukhtar Hossain:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Md Abul Kalam Azad:** Visualization, Validation, Resources, Investigation, Data curation. **Edward Binod Soren:** Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Md Nurul Alam:** Resources, Investigation, Data curation, Conceptualization. **Md Sharif Ahmed:** Writing – review & editing, Resources, Data curation, Conceptualization. **Md Saidul Islam:** Writing – original draft, Software, Methodology, Data curation. **Md Abdul Kaium:** Writing – review & editing, Methodology, Data curation. **Tabaraka Binte Ali:** Software, Methodology, Data curation.

**Fouzia Akter:** Writing – original draft, Software, Methodology, Data curation. **Md Imran Ali:** Visualization, Software, Investigation, Conceptualization. **Sirajam Monira:** Software, Methodology, Formal analysis, Data curation.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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